

BUNDABERG, AustraliaWMO: **943870**Lat: **24.9069S**Lon: **152.3220E**Elev: **32**StdP: **100.94**Time Zone: **10.00 (E10)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	9.0	-2.6	3.1	16.9	-0.4	3.7	16.5	10.0	19.4	9.1	18.8	2.4	230	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.6	31.7	23.6	30.8	23.3	30.0	22.9	25.7	28.9	25.2	28.3	24.6	27.8	5.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.9	20.0	27.4	24.3	19.3	27.0	23.7	18.6	26.6	79.7	28.9	77.4	28.4	75.0	27.8	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	9.1	8.2	DB	4.9	34.8	0.9	1.2	4.2	35.7	3.7	36.4	3.1	37.1	2.5	38.0	(4)
(5)			WB	3.0	26.9	0.8	0.9	2.5	27.5	2.0	28.1	1.6	28.6	1.0	29.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.5	25.8	25.6	24.6	22.3	19.5	17.2	16.4	17.2	19.7	21.8	23.4	25.0	(6)	
	DBStd	3.76	1.47	1.54	1.37	1.55	1.95	2.15	1.95	1.96	1.85	1.86	1.78	1.81	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	173	0	0	0	1	12	46	63	44	6	1	0	0	(9)	
	CDD10.0	4205	489	437	453	371	293	214	199	224	292	366	401	465	(10)	
	CDD18.3	1336	231	203	195	121	47	11	5	10	48	108	151	207	(11)	
	CDH23.3	9149	1974	1646	1334	596	156	21	16	48	228	546	939	1643	(12)	
	CDH26.7	2176	586	488	320	74	5	0	0	1	18	64	173	447	(13)	
(14) Wind	WSAvg	4.1	4.7	4.5	4.3	4.1	3.7	3.6	3.5	3.7	4.0	4.3	4.3	4.5	(14)	
(15) Precipitation	PrecAvg	1015	175	155	108	61	73	48	39	36	35	73	92	128	(15)	
	PrecMax	1892	693	729	432	277	382	357	382	135	146	243	227	480	(16)	
	PrecMin	483	1	14	7	3	10	0	0	0	0	10	3	7	(17)	
	PrecStd	328	140	138	96	57	70	66	59	33	36	55	57	105	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.6	32.9	31.8	29.8	27.3	25.2	25.0	26.1	28.7	29.7	31.2	33.1	(19)	
		MCWB	24.8	24.4	23.5	21.9	19.2	18.6	16.9	17.4	18.3	19.6	20.6	21.8	(20)	
	2%	DB	31.3	31.4	30.5	28.4	26.0	23.8	23.6	24.6	26.8	28.2	29.5	31.0	(21)	
		MCWB	24.0	23.9	23.1	21.0	18.7	17.6	16.2	16.8	18.3	20.4	21.7	23.4	(22)	
	5%	DB	30.5	30.5	29.5	27.5	25.0	22.9	22.5	23.6	25.6	27.2	28.5	29.9	(23)	
		MCWB	23.5	23.5	22.5	20.4	18.3	17.1	15.8	16.1	18.0	20.1	21.2	22.7	(24)	
	10%	DB	29.6	29.6	28.6	26.6	24.1	21.8	21.5	22.5	24.7	26.2	27.6	29.1	(25)	
		MCWB	23.0	23.0	22.1	20.1	17.9	16.4	15.3	15.5	17.6	19.4	20.7	22.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.6	26.1	25.8	24.5	22.4	20.6	20.3	20.1	22.0	23.2	24.6	26.1	(27)	
		MCDB	30.0	29.6	28.4	26.9	24.6	22.6	22.5	23.5	25.0	26.5	27.9	29.7	(28)	
	2%	WB	25.8	25.5	24.8	23.0	21.3	19.5	18.9	19.1	20.8	22.5	23.6	25.2	(29)	
		MCDB	29.0	28.9	27.7	25.7	23.5	21.6	21.1	22.1	24.2	25.7	27.1	28.5	(30)	
	5%	WB	25.2	24.9	24.1	22.1	20.2	18.6	17.7	18.1	20.1	21.7	22.7	24.4	(31)	
		MCDB	28.3	28.3	27.3	25.4	23.2	21.1	20.2	21.2	23.2	25.0	26.3	27.8	(32)	
	10%	WB	24.6	24.4	23.5	21.3	19.2	17.7	16.7	17.2	19.4	21.0	22.1	23.7	(33)	
		MCDB	27.8	27.7	26.7	24.9	22.5	20.7	19.8	20.5	22.5	24.3	25.7	27.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	7.6	8.0	8.7	9.6	9.5	10.7	11.2	10.4	9.0	8.3	7.8	(35)	
		MCDBR	8.4	8.8	9.1	9.6	10.5	10.1	11.6	12.1	11.4	9.7	9.1	8.6	(36)	
	5% WB	MCWBR	2.7	2.9	3.1	3.6	4.5	4.8	5.5	5.6	4.7	3.7	3.3	3.0	(37)	
		MCWBR	7.0	7.3	7.3	8.0	8.3	8.1	8.8	9.7	8.9	7.8	7.7	7.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.391	0.384	0.371	0.343	0.322	0.315	0.306	0.318	0.352	0.376	0.378	0.392	(40)	
		taud	2.513	2.543	2.567	2.635	2.639	2.653	2.656	2.609	2.506	2.470	2.499	2.491	(41)	
	Ebn at Noon		953	945	930	914	894	880	903	927	933	941	959	953	(42)	
		Edh at Noon	114	108	101	87	79	74	76	87	105	115	115	117	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.64	6.05	5.34	4.70	3.94	3.39	3.85	4.74	5.75	6.30	6.78	6.79	(44)	
	RadStd		0.48	0.56	0.45	0.36	0.20	0.26	0.30	0.34	0.42	0.49	0.63	0.69	(45)	

Historical Trends

		DBAvg	Heating			Cooling
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